

Classification, Regression, Prediction, and Inference

An Analysis of Three Statistical-Learning Problems

Abstract

Statistical learning provides a framework for analysing relationships between variables and for making predictions from observed data. A fundamental distinction in statistical learning is between regression and classification, which is determined primarily by the nature of the response variable. Regression problems involve quantitative responses, whereas classification problems involve qualitative or categorical responses. Another important distinction concerns the purpose of an analysis. A study may primarily seek inference, where the objective is to understand the relationship between predictors and a response, or prediction, where the objective is to predict the response for a new observation. This assignment applies these concepts to three practical scenarios involving CEO salary, product success or failure, and changes in the USD/Euro exchange rate. For each scenario, the response variable, predictors, problem type, analytical objective, number of observations (n), and number of predictors (p) are identified and justified. The analysis demonstrates that scenario (a) is a regression problem primarily concerned with inference, scenario (b) is a classification problem primarily concerned with prediction, and scenario (c) is a regression problem primarily concerned with prediction.

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1 Introduction

Statistical learning is an important area of modern statistics that focuses on understanding patterns and relationships in data and using those relationships to make predictions. In supervised learning, a researcher observes a response variable, denoted by Y , together with one or more predictor variables, denoted by $X_1, \dots, X_p$. The objective may be to understand how the predictors are related to the response or to use the predictors to obtain accurate predictions for future observations. James et al. (2021) identify regression and classification as two fundamental types of supervised-learning problems. The distinction between them is primarily based on the nature of the response variable: regression concerns quantitative responses, while classification concerns qualitative or categorical responses.

In addition to distinguishing regression from classification, it is important to distinguish between *inference* and *prediction*. Inference focuses on understanding the relationships between the predictors and the response. For example, a researcher may want to determine whether company profit is associated with CEO salary. Prediction, in contrast, focuses on accurately predicting the response for a new observation. For example, a company may want to predict whether a newly introduced product will succeed or fail based on information about similar products launched in the past.

These distinctions are important because the same statistical-learning techniques may be used for different purposes, but the interpretation of the results depends on the research objective. A model designed primarily for prediction may emphasize predictive accuracy, whereas an analysis designed for inference may emphasize the estimated relationships between variables and the interpretation of model coefficients (James et al., 2021).

This assignment examines three scenarios. The first concerns the factors associated with CEO salary among the top 500 firms in the United States. The second concerns predicting whether a new product will be a success or failure. The third concerns predicting percentage changes in the USD/Euro exchange rate using weekly changes in stock markets. For each scenario, the response variable and predictors are identified, the problem is classified as either regression or classification, the main objective is identified as either inference or prediction, and the values of n and p are determined.

2 Understanding the Four Fundamental Concepts

2.1 Regression

A **regression problem** occurs when the response or output variable Y is quantitative or numerical. The objective is generally to estimate or predict a numerical value using one or more predictor variables. Linear regression is one of the fundamental statistical-learning methods for modelling quantitative responses (James et al., 2021).

Examples of regression problems include:

- Predicting a salary of \$100,000;
- Predicting a house price of \$350,000;
- Predicting a temperature of 25°C;
- Predicting an exchange-rate change of 0.8%.

A multiple linear regression model can generally be represented as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon, \quad (1)$$

where Y is the quantitative response variable, $X_1, \dots, X_p$ are the predictor variables, β_0 is the intercept, $\beta_1, \dots, \beta_p$ are regression coefficients, and ϵ represents the random error term.

For example, if CEO salary is being modelled using profit and number of employees, salary is the response variable and the other variables are predictors. Since salary is measured numerically, the problem is a regression problem.

2.2 Classification

A **classification problem** occurs when the response variable is qualitative or categorical. Rather than predicting a numerical value, the objective is to assign an observation to one of a set of predefined categories (James et al., 2021).

Examples of classification problems include:

- Success versus failure;
- Yes versus no;
- Disease versus no disease;
- Spam versus not spam.

For example, suppose a company wants to determine whether a new product will succeed. A classification model could estimate the probability of success. If the estimated probability of success is 0.82, the product may be classified as a likely success according to a specified classification rule.

The important distinction is that the response variable in classification is categorical. In contrast, a regression model would predict a numerical outcome such as sales revenue, profit, or market value.

2.3 Prediction

The primary objective of **prediction** is to obtain an accurate estimate of Y for a new observation. In a prediction problem, the researcher is mainly interested in how accurately a fitted model can predict outcomes that were not used to estimate the model.

For example, consider a company that is planning to introduce a new product. Suppose the company has information about:

- the price of the product;
- the marketing budget;
- the prices charged by competitors; and
- other characteristics of the product.

The company may ask:

Based on the characteristics of the new product and information from previously launched products, will the new product succeed?

This is a prediction problem because the main objective is to determine the likely outcome for a new observation.

Prediction does not necessarily require a detailed understanding of the causal mechanisms underlying the relationships among the variables. Instead, the primary concern is the accuracy of predictions for new or future observations (James et al., 2021).

2.4 Inference

The primary objective of **inference** is to understand the relationship between the predictors $X_1, \dots, X_p$ and the response Y . Researchers interested in inference may ask which predictors are associated with the response, whether particular predictors have statistically important relationships with the response, and how changes in predictors are related to changes in the response.

For example, suppose a researcher is investigating CEO salaries and wants to determine whether company profit, number of employees, and industry are associated with CEO compensation. The researcher might ask:

Does the number of employees have an important relationship with CEO salary?

This is an inference question because the objective is to understand the relationship between the explanatory variables and CEO salary rather than simply to predict the salary of a new CEO.

Therefore, a useful distinction is:

Prediction asks “How accurately can we predict Y ?”

whereas

Inference asks “How and why is Y related to the predictors?”

The distinction between prediction and inference is an important component of statistical learning (James et al., 2021).

3 Scenario (a): CEO Salary

3.1 Problem Statement

The first scenario states:

We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

The objective is to determine whether this is a regression or classification problem, whether the primary objective is inference or prediction, and the values of n and p .

3.2 Step 1: Identifying the Response Variable

The response variable is CEO salary. Therefore,

$$Y = \text{CEO salary.} \tag{2}$$

The predictors are:

$$X_1 = \text{Profit,} \tag{3}$$

$$X_2 = \text{Number of employees,} \tag{4}$$

$$X_3 = \text{Industry.} \tag{5}$$

Thus, CEO salary is the outcome that the researcher wants to explain, while profit, number of employees, and industry are explanatory variables.

3.3 Step 2: Regression or Classification?

CEO salary is a quantitative numerical variable. For example, one CEO may earn \$5 million per year, another may earn \$10 million, and another may earn \$20 million.

Because the response variable is numerical, the problem is a:

$$\boxed{\text{Regression problem.}} \quad (6)$$

A simplified regression model could be expressed as

$$\text{Salary} = \beta_0 + \beta_1(\text{Profit}) + \beta_2(\text{Employees}) + \beta_3(\text{Industry}) + \epsilon. \quad (7)$$

It is important to note that industry is a categorical predictor. In an actual regression analysis, industry would normally be represented using dummy or indicator variables. Consequently, if industry has several categories, the number of estimated regression coefficients associated with industry would exceed one. However, for the purpose of this textbook exercise, industry is counted as one predictor, resulting in $p = 3$.

3.4 Step 3: Prediction or Inference?

The scenario explicitly states:

“We are interested in understanding which factors affect CEO salary.”

The word “understanding” indicates that the main objective is to investigate the relationship between CEO salary and the explanatory variables.

The researcher is not primarily asking:

What will the salary of the CEO of a new company be?

Instead, the researcher wants to understand how factors such as profit, number of employees, and industry are associated with CEO salary.

Therefore, this is an:

$$\boxed{\text{Inference problem.}} \quad (8)$$

For example, an analysis could investigate whether firms with higher profits tend to pay their CEOs higher salaries after accounting for other observed characteristics. Such a result would contribute to understanding the relationship between profitability and CEO compensation.

It is worth noting that the wording “affect” should be interpreted carefully. Observational data alone generally establish statistical associations rather than definitive causal effects unless additional assumptions or research designs support causal interpretation.

3.5 Step 4: Determining n

The value n represents the number of observations in the dataset.

The study contains information on the top 500 firms. Therefore,

$$\boxed{n = 500}. \quad (9)$$

Each firm represents one observation.

3.6 Step 5: Determining p

The value p represents the number of predictor variables.

The predictors are:

1. Profit;
2. Number of employees;
3. Industry.

Therefore,

$$p = 3. \quad (10)$$

The CEO salary is not included in p because it is the response variable.

3.7 Answer for Scenario (a)

The final classification is:

$$\boxed{\text{Regression; Inference; } n = 500; \quad p = 3.} \quad (11)$$

3.8 Illustrative Example

Suppose the analysis finds that, after accounting for industry and the number of employees, firms with higher profits tend to have higher CEO salaries. This would be an inference finding because the researcher is attempting to understand the relationship between profitability and CEO compensation.

4 Scenario (b): New Product Success or Failure

4.1 Problem Statement

The second scenario states:

We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

4.2 Step 1: Identifying the Response Variable

The response variable is whether the product is successful or unsuccessful:

$$Y = \text{Success or Failure.} \quad (12)$$

There are two possible response categories:

- Success;
- Failure.

Therefore, the response is categorical.

4.3 Step 2: Regression or Classification?

Because the response variable is categorical rather than numerical, this is a:

$$\boxed{\text{Classification problem.}} \quad (13)$$

For example, a classification model could estimate:

$$P(\text{Success} \mid X) = 0.82. \quad (14)$$

This means that, given the observed predictor variables, the model estimates an 82% probability of success.

A classification rule could then classify the product as a success if the estimated probability exceeds a specified threshold. A threshold of 0.50 is a simple example, although the appropriate threshold depends on the costs associated with incorrect classifications.

This differs from regression. A regression model might predict that a product will generate \$10 million in sales, whereas a classification model predicts a category such as “success” or “failure.”

4.4 Step 3: Prediction or Inference?

The problem states that the company is considering launching a **new product** and wants to know whether it will be successful.

The objective is therefore to predict the outcome of a new observation.

Hence, the problem is primarily concerned with:

$$\boxed{\text{Prediction.}} \tag{15}$$

For example, suppose the proposed product has:

- Price = \$50;
- Marketing budget = \$2 million;
- Competition price = \$45;
- Ten additional characteristics.

A classification model could use information from the 20 previously launched products to predict whether the proposed product is likely to succeed.

If the model estimates an 85% probability of success, management could use this information, together with other business considerations, when deciding whether to launch the product.

4.5 Step 4: Determining n

The data contain information on 20 previously launched products.

Therefore,

$$\boxed{n = 20}. \tag{16}$$

Each product represents one observation.

4.6 Step 5: Determining p

The predictors explicitly identified are:

1. Price charged;
2. Marketing budget;
3. Competition price;
4. Ten additional variables.

Therefore,

$$p = 3 + 10 = 13. \tag{17}$$

Hence,

$$\boxed{p = 13}. \quad (18)$$

The success/failure outcome is not included in p because it is the response variable Y .

4.7 Answer for Scenario (b)

The final classification is:

$$\boxed{\text{Classification; Prediction; } n = 20; \quad p = 13.} \quad (19)$$

4.8 Illustrative Example

Suppose a classification model estimates an 85% probability that the new product will succeed. If the decision threshold is 50%, the model would classify the product as a likely success.

The purpose is predictive rather than inferential because the principal question concerns the likely outcome of a new product rather than understanding the individual effects of the predictors.

5 Scenario (c): USD/Euro Exchange Rate

5.1 Problem Statement

The third scenario states:

We are interested in predicting the percentage change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence, we collect weekly data for all of 2012. For each week we record the percentage change in the USD/Euro, the percentage change in the US market, the percentage change in the British market, and the percentage change in the German market.

5.2 Step 1: Identifying the Response Variable

The response variable is the percentage change in the USD/Euro exchange rate:

$$Y = \text{percentage change in the USD/Euro exchange rate.} \quad (20)$$

This is a quantitative numerical variable.

For example, the exchange rate might change by:

- +0.5%;
- -1.2%;
- +0.8%.

Since these values are numerical, the response is quantitative.

5.3 Step 2: Regression or Classification?

Because the response variable is numerical, the problem is a:

$$\boxed{\text{Regression problem.}} \quad (21)$$

A possible multiple regression model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon, \quad (22)$$

where:

$$Y = \text{percentage change in the USD/Euro exchange rate}, \quad (23)$$

$$X_1 = \text{percentage change in the US market}, \quad (24)$$

$$X_2 = \text{percentage change in the British market}, \quad (25)$$

$$X_3 = \text{percentage change in the German market}. \quad (26)$$

The regression model attempts to relate weekly changes in the three stock markets to weekly changes in the USD/Euro exchange rate.

5.4 Step 3: Prediction or Inference?

The scenario explicitly states:

“We are interested in predicting the percentage change in the USD/Euro exchange rate.”

The objective is therefore prediction.

Hence,

$$\boxed{\text{Prediction.}} \quad (27)$$

For example, if the US, British, and German stock markets all experience positive weekly changes, the fitted model could be used to estimate the expected change in the USD/Euro exchange rate.

Suppose the model produces a predicted exchange-rate change of +0.4% for a particular combination of stock-market movements. This is an example of using regression for prediction.

5.5 Step 4: Determining n

The data are collected weekly throughout 2012. A year contains approximately 52 weeks. Therefore,

$$\boxed{n = 52}. \quad (28)$$

Each week represents one observation.

5.6 Step 5: Determining p

The predictors are:

1. Percentage change in the US market;
2. Percentage change in the British market;
3. Percentage change in the German market.

Therefore,

$$\boxed{p = 3}. \quad (29)$$

The percentage change in the USD/Euro exchange rate is the response and therefore is not counted among the predictors.

5.7 Answer for Scenario (c)

The final classification is:

$$\boxed{\text{Regression; Prediction; } n = 52; \quad p = 3.} \quad (30)$$

6 Comparative Summary

The results for the three scenarios are summarized in Table 1.

The table demonstrates the importance of identifying both the nature of the response variable and the purpose of the analysis. Scenario (a) has a numerical response and therefore uses regression, while its objective is inference. Scenario (b) has a categorical response and therefore uses classification, while its objective is prediction. Scenario (c) has a numerical response and therefore uses regression, while its objective is also prediction.

Table 1: Classification of the Three Statistical-Learning Problems

Part	Response Y	Predictors X	Problem Type	Objective	n	p
(a)	CEO salary	Profit, employees, industry	Regression	Inference	500	3
(b)	Success/failure	Price, marketing budget, competi- tion price, and 10 other variables	Classification	Prediction	20	13
(c)	Percentage change in USD/Euro	US, British, and German market changes	Regression	Prediction	52	3

7 Interpretation of n and p

Two quantities that are important in describing a statistical-learning dataset are n and p . The quantity n represents the number of observations, while p represents the number of predictor variables.

In scenario (a), there are 500 firms, so $n = 500$. The three predictors are profit, number of employees, and industry, giving $p = 3$.

In scenario (b), there are 20 previously launched products, so $n = 20$. The three explicitly identified predictors are price, marketing budget, and competition price, together with ten additional variables. Therefore, $p = 13$.

In scenario (c), the data are collected weekly for approximately 52 weeks, giving $n = 52$. There are three predictors: changes in the US, British, and German stock markets. Hence, $p = 3$.

It is important to remember that the response variable Y is not included in the count of p . Thus, if a dataset contains one response and three predictors, then $p = 3$, not $p = 4$.

8 Discussion

The three scenarios demonstrate that identifying the response variable is the first important step when determining the type of statistical-learning problem. In scenarios (a) and (c), the response variables are quantitative. CEO salary is measured numerically, while the USD/Euro exchange-rate change is expressed as a percentage. Consequently, both are regression problems.

In contrast, scenario (b) has a categorical response consisting of two possible outcomes: success or failure. Therefore, it is a classification problem.

The second important consideration is the purpose of the analysis. Scenario (a) focuses

on understanding which factors are associated with CEO salary. This makes it primarily an inference problem. In scenarios (b) and (c), the stated objective is to make predictions for new or future observations, making them prediction problems.

There is also an important distinction between statistical association and causation. In scenario (a), the wording of the question refers to factors that “affect” CEO salary. However, simply observing that profit and CEO salary are associated does not necessarily prove that profit causes changes in CEO salary. Other variables may influence both profit and salary, and observational data may contain confounding factors. Therefore, a careful statistical interpretation would normally use terms such as “associated with” or “related to” unless the research design supports causal conclusions.

Furthermore, the values of n and p provide useful information about the structure of the data. Scenario (b), for example, has only 20 observations but 13 predictors. This represents a relatively small sample compared with the number of predictors and could create challenges for model estimation and predictive performance. In contrast, scenario (a) has 500 observations and only three predictors as counted in the exercise, giving substantially more observations relative to the number of predictors.

9 Conclusion

This assignment has examined three statistical-learning problems using the fundamental distinctions between regression, classification, prediction, and inference. The analysis demonstrates that the classification of a statistical problem depends first on the nature of the response variable and second on the purpose for which the model is being developed.

Scenario (a), involving CEO salaries among the top 500 firms in the United States, is a **regression problem** because CEO salary is a quantitative response variable. The principal objective is **inference** because the researcher wants to understand which factors are associated with CEO salary. There are 500 observations and three predictors, giving $n = 500$ and $p = 3$.

Scenario (b), involving the success or failure of a new product, is a **classification problem** because the response variable is categorical. The primary objective is **prediction** because the company wants to determine whether a new product is likely to succeed or fail. There are 20 observations and 13 predictors, giving $n = 20$ and $p = 13$.

Scenario (c), involving the percentage change in the USD/Euro exchange rate, is a **regression problem** because the response is quantitative. The principal objective is **prediction** because the stated goal is to predict exchange-rate changes based on weekly changes in stock markets. There are approximately 52 weekly observations and three predictors, giving $n = 52$ and $p = 3$.

The central lesson is therefore that **regression versus classification is determined primarily by the type of response variable, whereas inference versus prediction**

is determined primarily by the objective of the analysis. Quantitative responses generally lead to regression problems, while categorical responses lead to classification problems. Understanding these distinctions is fundamental to selecting appropriate statistical-learning techniques and interpreting their results correctly.

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